

Andrew DeLaCruz

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EDUCATION

University of California, Davis

Davis, CA

Bachelor of Science in Mechanical Engineering, GPA: 3.30/4.00

Expected June 2029

- **Completed:** Engineering Programming (MATLAB) **In Progress (by Jun 2027):** Statics, Dynamics, Thermodynamics, Engineering Graphics (SolidWorks), Circuits I & II, Properties of Materials

EXPERIENCE

Cogwell, Inc. Santa Clara, CA

Jun 2025 – Sep 2025

Calibration & Test Engineering Intern

Jun 2026 – Sep 2026

- Developed automated calibration procedures in Fluke MET/CAL and Python test scripts for 9 instrument models, including Keysight spectrum analyzers, Fluke calibrators, and Tektronix oscilloscopes
- Performed root cause analysis of PCB faults using schematics; repaired boards through solder rework
- Executed verification and validation (V&V) procedures against calibration standards using multimeters, oscilloscopes, spectrum analyzers, and signal generators
- Calibrated 30+ precision instruments to SOPs and documented results in calibration reports

PROJECTS

Nightstand Hub | SolidWorks, FDM 3D Printing, ESP32, Mechatronics

Aug 2026 – Present

- Designed a three-part, $11 \times 8 \times 16$ cm SolidWorks enclosure for two displays, an ESP32 microcontroller, a speaker, and a temperature sensor, dimensioned with calipers and micrometers; delivered a working prototype for \$31.72 in parts
- Engineered an adhesive-free assembly with press fits and locating pegs, reaching a working fit in 2 print iterations (DFM); documented critical fits in GD&T drawings using datums and geometric controls
- Thermally isolated the temperature sensor from heat-generating electronics with a tapered top and rear-panel airflow path, verifying readings against an external thermometer
- Traced a full display failure to a mis-held GPIO pin through pull-up/pull-down testing with a multimeter; rewired each display and sensor onto a dedicated bus, eliminating a recurring intermittent connection fault
- Developed firmware with AI-assisted tools (Claude Code), integrating weather, calendar (iCal), and transit APIs into a two-screen touchscreen UI; validated each subsystem on hardware, now in daily use

FlashBack: 270° Panoramic Photo Booth | SolidWorks, Python, OpenCV, Raspberry Pi

Mar 2026 – May 2026

- Co-designed a battery-powered photo booth with a 22-person team in 6 weeks, delivering a field-tested \$339.60 prototype for DUEN (Davis Undergrad Engineering Network) events; served on the EECS team and as the mechanical liaison
- Redesigned the load path with a ball-bearing base that took chassis weight off the servo shaft, and selected counterweights that kept the booth stable with the mast extended and eliminated motor-induced rattle
- Designed electromechanical packaging and cable routing for a Raspberry Pi 4, battery, buck converters, fuse box, and thermal printer; calculated the power budget to support 6-hour battery runtime at events
- Developed a Python/OpenCV stitching pipeline for 16 overlapping images, tuning overlap, exposure locking, and PWM servo settle time to turn unreliable stitching into consistent 270° panoramas; built a Tkinter UI
- Assembled and tested the booth (heat-set inserts, soldering, warped-print fixes); traced a Raspberry Pi short circuit minutes before technical review and swapped in a backup, keeping the demo on schedule

Fold or Roll | MATLAB, App Designer, ThingSpeak, REST API

Jan 2026 – Mar 2026

- Built a two-player MATLAB game synced across two computers in near real time via ThingSpeak's IoT REST API
- Designed a compressed JSON data format and turn-gating logic that kept each player's dice private

TECHNICAL SKILLS

CAD & Fabrication: SolidWorks, engineering drawings, DFM, FDM 3D printing, heat-set inserts, soldering, calipers

Programming: Python, MATLAB, Fluke MET/CAL, OpenCV, Git, Claude Code

Electronics & Test Equipment: Oscilloscopes, spectrum analyzers, multimeters, signal generators, PCB rework, ESP32, Raspberry Pi

Languages: English, Spanish